

WHAT DOES RETRIEVED MEMORY BUY? ACCURACY, EVIDENCE AND WRITE COST

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ABSTRACT

Agent memory can reduce information loss under a restricted prompt budget, but query-only accounting may conceal its construction cost. We audit file-prefix memory and retrieved working memory in JiuwenSwarm using the same qwen-plus model, fact episodes and full-section character limits. The study covers 20 adapted LongMemEval questions at two budgets, with isolated query histories and one recorded run per budget. At the restricted budget, answer accuracy rises from 0.0% to 65.0%; at the expanded budget it rises from 60.0% to 95.0%. In the retrieved arm, complete-evidence recall agrees with answer correctness for every observed question. However, model-backed memory writes account for most of its token expenditure. Its total cost is 21.70 times the baseline at the restricted budget and 15.64 times at the expanded budget, despite lower query-token counts. We provide the execution protocol, paired outcomes, descriptive uncertainty intervals and question-level diagnostics needed to interpret that trade-off. Construction, formatting and selection differ jointly between arms, so the observed advantage cannot be attributed to ranking alone. The results describe a selected single-session slice, not an official benchmark score or an evaluation of persistent multi-agent memory.

1 INTRODUCTION

A language-model agent can retain project facts in a file and expose a prefix of that file to its next question. This is inexpensive to construct, but a fixed prompt allowance can remove relevant facts before the model sees them. An alternative is to store exchanges as separate memory items and select a subset for each question. That design introduces a second resource decision: how the items are constructed. If construction itself invokes a model, measuring only the final query can give an incomplete account of operational cost. A useful comparison therefore needs to follow the information delivered to the query and the model calls required to prepare it.

This paper presents an auditable systems study of that trade-off. The research question is: how do file-based project memory and hybrid retrieval compare in answer accuracy and total model cost under equal context-character budgets? Three sub-questions structure the investigation. RQ1 asks whether accuracy differs between the two arms when both receive the same character budget. RQ2 asks whether complete evidence recall is associated with correct answers, and whether partial evidence can suffice. RQ3 asks how total token cost, including memory-write calls, compares between the arms.

The study audits an existing implementation rather than proposing a new retrieval algorithm. Both arms receive identical facts, questions, model identity, and system prompt; only the memory-construction and selection protocol differ. The baseline writes facts to an isolated JI-UWENSWARM.md file and presents a prefix-truncated view. The enhanced arm writes completed user-assistant exchanges to process-local L1 memory through model-backed conversations and selects items using a fixed lexical ranking. Neither arm employs learned embeddings, reflection modules, or external tools.

The study contributes an auditable comparison, a cost decomposition and a question-level diagnostic record. The comparison pairs identical questions under restricted and expanded memory budgets instead of treating the two budgets as independent datasets. The decomposition separates query

tokens from model-backed writes, allowing an accuracy advantage to be assessed alongside its construction expense. The diagnostics distinguish missing complete evidence from an incorrect answer and expose a case where a partial fact still contains an accepted answer. These contributions concern measurement and implementation behaviour; they do not establish a new retrieval algorithm or state-of-the-art performance.

The scope is deliberately narrow. The experiments do not validate an end-to-end paper-generation pipeline, do not test multi-session reasoning or temporal updates, and do not compare against published memory systems. The adapted LongMemEval slice covers only single-session factual extraction. All reported intervals are descriptive Wilson intervals over the twenty selected questions; no significance test is claimed or justified by the single-run protocol.

2 RELATED WORK

Memory control and organisation for language-model agents has attracted several architectural proposals. MemGPT introduces an operating-system-inspired virtual context manager that moves information across memory tiers, demonstrating document analysis and multi-session chat scenarios (Packer et al., 2023). The present study’s fixed L1 Rail does not implement MemGPT’s autonomous virtual-memory controller; MemGPT serves only as conceptual context for the tiered-storage vocabulary. A-MEM creates structured notes and dynamically links and evolves memories following a Zettelkasten-inspired organisation (Xu et al., 2025). Our fixed lexical ranking lacks autonomous linking and evolution, and no empirical comparison to A-MEM was run. Generative Agents synthesise natural-language experience records into reflections and dynamically retrieve them for planning behaviour in an interactive simulation (Park et al., 2023). This study neither implements reflection or planning nor tests social behaviour; it draws only on the general notion of retrieving stored records at query time.

Evidence access and position effects constitute a second theme. Lost in the Middle demonstrates that relevant information position affects multi-document question answering and key-value retrieval in long contexts, with middle positions often harder to exploit (Liu et al., 2023). That finding motivates attention to how information is placed within a prompt, but it does not establish that the prefix truncation observed in our baseline is caused by the same attentional phenomenon; the baseline failure is deterministic removal of text beyond the budget, not degraded attention to retained text. MemoryBank supports retrieval and updates for sustained personalised interaction, modelling forgetting and reinforcement influenced by elapsed time and importance (Zhong et al., 2023). Our temporal-decay term is a simpler heuristic and does not replicate MemoryBank’s companion evaluation or its reinforcement dynamics.

Evaluation of long-term interactive memory forms the third cluster. LongMemEval benchmarks chat assistants on extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention, explicitly separating indexing, retrieval, and reading stages (Wu et al., 2024). The submitted adaptation covers only a narrow single-session-user extraction slice converted into fixed fact episodes; it does not exercise multi-session reasoning, temporal reasoning, updates, or abstention, and the resulting scores are not official LongMemEval benchmark results. Together, these six sources frame the design space within which the present two-arm comparison sits, while none is replicated or empirically benchmarked here.

3 METHOD

The method diagram contrasts the two executed paths from a shared fact episode to a fresh query conversation and a paired audit. The baseline constructs a file-derived section and retains its prefix. The retrieved arm constructs dialogue-shaped memory items, ranks them, and packs complete formatted items into the allowed section. The same model answers both queries, but construction and presentation differ jointly. Accordingly, the estimand is the observed difference between these operational systems under a matched section budget, rather than the isolated effect of lexical retrieval.

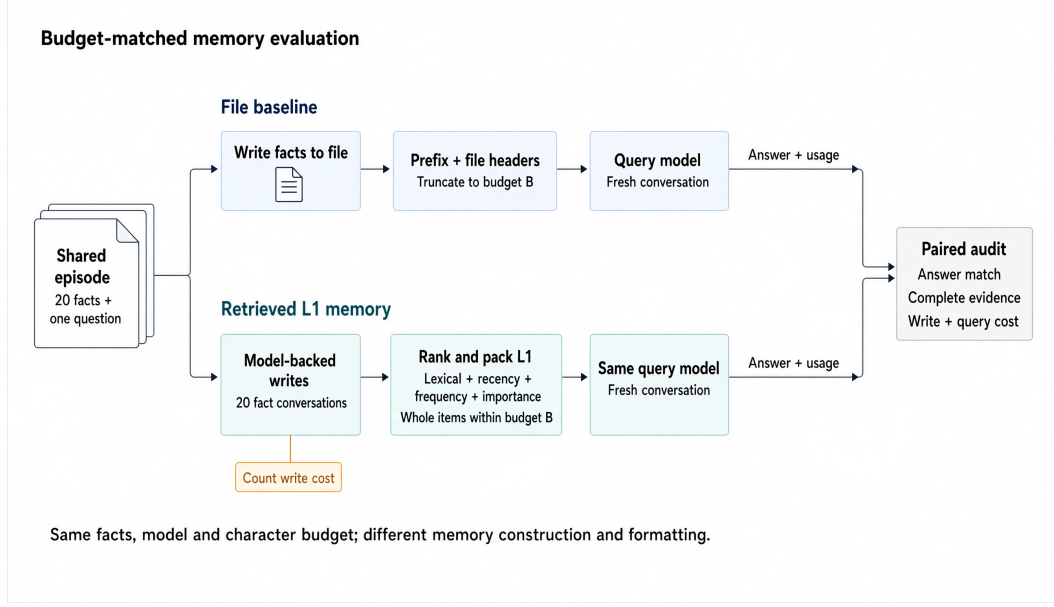


Figure 1: Budget-matched evaluation of two operational memory paths. The retrieved arm incurs model-backed write cost before its isolated query. Equal character caps include each arm’s own formatting overhead. This schematic illustrates the audited implementation; it is not an experimental result.

3.1 FILE-BASED PROJECT MEMORY

The baseline uses ProjectMemoryRail. Facts are written directly to an isolated JI-UWENSWARM.md file without model-backed seeding. At query time, the runner reads the file, includes its filename and modification metadata, and truncates the complete memory section to the allowed prefix. All of these characters count against the budget. There is no query-dependent re-ordering: later facts can be removed regardless of relevance, and the truncation boundary may split a fact. Consequently, the arm is a concrete file-prefix baseline rather than an embedding retriever. If complete evidence is absent, that establishes what the model did not receive under this protocol; it does not establish how a differently ordered or reformatted file would perform.

3.2 HYBRID SELECTION FROM WORKING MEMORY

The enhanced arm uses EnhancedMemoryRail to store completed user-assistant exchanges in process-local L1 memory. Each fact is supplied through a separate model-backed conversation, so the resulting item contains both the supplied text and an assistant response. Retrieval combines Jaccard overlap of token sets with recency, access frequency and a heuristic importance score. The formalization below gives the audited default weights and normalization; these coefficients were not fitted on the evaluation questions. A fixed-size candidate set is ranked, each display item is individually capped, and a greedy packer adds complete formatted items while space remains. An item that does not fit is skipped, allowing a later shorter item to be considered. Selection and packing can therefore omit a fact for different reasons, even when total memory capacity is sufficient. This path invokes no embedding model and provides no cross-restart persistence. The implementation defines other memory layers, but they are not exercised by this Rail. MemGPT supplies conceptual context for tiered-memory terminology (Packer et al., 2023); its autonomous context controller is not implemented here.

3.3 ISOLATION AND COST ACCOUNTING

Each public-data question receives its own fact episode and fresh agent instances. For the baseline, facts are written directly to disk. For the enhanced arm, the runner conducts model-backed memory-

writing conversations: each fact episode is presented as a user turn, the model produces an assistant acknowledgement, and the completed exchange is stored in L1. This seeding phase requires 400 model calls per budget and is the source of the write-cost asymmetry. The formal question then uses a new conversation ID so that no seeding dialogue leaks into the query context. The runner verifies isolated query history and confirms actual memory injection before scoring. Arm execution order is alternated across questions to control for potential API-level rate-limiting or temporal drift during the run; because agents are instantiated fresh per question, there is no cross-question shared state contamination to mitigate. Both query-time token usage and memory-write token usage are recorded separately. Latency measurements include the write phase and are amortised over the paired observations. Correctness is assessed by accepted-answer string matching against a predefined answer list; evidence recall requires the complete target fact to survive prompt truncation and appear in the injected context. The evaluation design adapts the single-session extraction paradigm of LongMemEval (Wu et al., 2024), though the resulting slice is not the official benchmark.

3.4 BUDGET AND COST FORMALIZATION

Let q be a question, M its episode memory, and B the cap on the entire injected memory section, including headings and metadata. The baseline constructs a file-derived section and keeps its prefix. The retrieved arm scores stored exchanges, takes at most $k = 10$ candidates, and greedily appends complete formatted items that fit. An over-budget item is skipped rather than partially appended; later smaller items can still be considered. Thus equal B does not imply equal usable fact content or identical prompt formatting.

$$s(q, m) = 0.4J(T(q), T(m)) + 0.32^{-a(m)/h} + 0.2 \min\left\{\frac{\log(1 + f(m))}{\log(101)}, 1\right\} + 0.1I(m), \quad (1)$$

where T lowercases text and keeps alphanumeric/underscore word tokens longer than one character; $J(A, C) = |A \cap C| / |A \cup C|$ (zero for an empty set); $a(m)$ is age in seconds; $h = 86400$ is the decay half-life; $f(m)$ is the recorded access count; and $I(m)$ is a bounded heuristic importance score. These are the implementation’s default weights, not learned or tuned coefficients. No embedding model is invoked. L1 holds at most 20 exchanges with a one-hour time-to-live; individual displayed exchanges are capped at 500 characters before packing.

The cost estimand includes initialization through model-backed memory writes:

$$C_g = \frac{\sum_{r \in \mathcal{Q}_g} u_r + \sum_{w \in \mathcal{W}_g} u_w}{N_{\text{valid}}}, \quad L_g = \frac{\sum_{r \in \mathcal{Q}_g} t_r + \sum_{w \in \mathcal{W}_g} t_w}{N_{\text{valid}}}. \quad (2)$$

Here u and t denote recorded token usage and elapsed time, respectively; $\mathcal{Q}_g$ and $\mathcal{W}_g$ include all recorded query and write attempts. The denominator counts valid matched question pairs. These timing sums exclude unrecorded process initialization and are not a full application wall-clock time. Missing usage remains unknown rather than zero.

For each valid pair, let $y_{g,i}$ indicate an accepted-answer substring match and $e_{g,i}$ indicate that the complete target fact survived in the injected text:

$$A_g = \frac{1}{N} \sum_i y_{g,i}, \quad R_g = \frac{1}{N} \sum_i e_{g,i}, \quad \Delta A = A_{\text{retrieved}} - A_{\text{file}}. \quad (3)$$

The two indicators differ: a truncated fact can contain a sufficient answer span. Wilson intervals describe finite-question uncertainty only. They do not estimate model-run variability, correct the selected slice’s bias, or establish significance.

Table 1: Audited execution protocol. No component ablation or weight tuning is claimed.

Step	Operation
Prepare	Create fresh isolated agents; give both arms the same fact episode.
Construct	Write the baseline file; seed retrieved L1 through separate fact conversations.
Query	Use a fresh conversation, no tools, one model iteration, and the assigned section budget.
Audit	Check actual prompt injection and isolated history before accepting a pair.
Aggregate	Match question IDs; count writes once per run; preserve excluded attempts in costs.

4 EXPERIMENTAL SETUP

The evaluation uses a selected LongMemEval-S single-session-user slice converted to fixed episodes containing one target answer turn and distractors. The two budgets share the same 20 questions and must not be counted as independent additional questions. Every episode supplies the configured number of facts, which matches the working-store capacity in the audited implementation. The recorded protocol therefore studies short-lived memory within that capacity rather than a workload designed to test eviction. All query responses use qwen-plus, no external tools and one model iteration. We report answer matching, complete-evidence recall, query and write token totals, query latency, and recorded latency including writes. The latter is an accounting sum, not the wall-clock duration of the whole application. There is no semantic judge or repeated-run estimate; both restrictions matter when interpreting close outcomes or timing differences.

4.1 RESTRICTED CONTEXT BUDGET

The restricted-budget condition allocates 600 characters to the full memory section. There are 20 unique questions, 1 repeat per question, yielding 20 paired observations. At this budget the baseline’s prefix-truncation policy structurally disadvantages it, as defined in the method section: the metadata consumes a fixed portion of the allowance, and the deterministic prefix order means that later facts in the serialised file are systematically excluded. The enhanced arm’s whole-item packing can select relevant items from any position in the store, provided each fits within the remaining budget. Metrics recorded include answer accuracy, complete evidence recall, total tokens (query plus seed writes), tokens per question, query latency in milliseconds, and end-to-end latency including the write phase. The small budget is the condition under which the prefix-truncation confound is most acute, and results at this budget should be interpreted with that structural asymmetry in mind.

4.2 EXPANDED CONTEXT BUDGET

The expanded-budget condition allocates 2400 characters to the full memory section, using the same 20 questions with 1 repeat, producing 20 paired observations. At this budget the baseline can accommodate more of the serialised file, reducing the fraction of facts lost to truncation. However, prefix order still determines which facts survive: facts appearing early in the file are always retained, while those near the end remain vulnerable if the file exceeds the budget. The enhanced arm’s packing behaviour is unchanged in logic but benefits from the larger budget by fitting more whole items. The same metrics are recorded: answer accuracy, complete evidence recall, total tokens including writes, tokens per question, query latency, and end-to-end latency. The expanded budget tests whether the accuracy gap observed at the restricted budget persists when the baseline is no longer structurally starved of context space.

5 RESULTS AND ANALYSIS

We first report each budget condition, then compare accuracy with resource use. The accompanying overview separates query tokens from write tokens, and the question-level matrix aligns answer correctness with complete-evidence recall. Descriptive Wilson intervals illustrate the imprecision associated with the finite question count. They neither correct the selected slice’s sampling bias

nor measure model stochasticity, and overlap or separation of these intervals is not used as a paired significance test.

5.1 RESTRICTED CONTEXT BUDGET

At the restricted budget of 600 characters, the baseline achieves accuracy of 0.0% and complete evidence recall of 0.0%, with 0 correct answers out of 20 questions. The enhanced rail achieves accuracy of 65.0% and evidence recall of 65.0%, with 13 correct out of 20. The accuracy gain is 65 percentage points. The paired decomposition shows 0 both-correct, 13 enhanced-only, 0 baseline-only, and 7 both-wrong outcomes. The baseline’s zero accuracy is consistent with deterministic prefix truncation removing every target fact from the prompt at this budget; all twenty baseline answers are the string UNKNOWN. The enhanced arm’s 7 failures have no complete target fact in the delivered context. The saved observations do not separate candidate-ranking failure, item truncation and packing exclusion, so they establish evidence absence without locating the failing stage. No enhanced answer was correct without full evidence (0 cases) and no enhanced answer was wrong despite full evidence (0 cases), indicating tight coupling between evidence presence and answer correctness in this arm.

Descriptive Wilson intervals for accuracy are 0.0% to 16.1% for the baseline and 43.3% to 81.9% for the enhanced arm. These intervals reflect question-sampling uncertainty over twenty items, not repeated-run variability. Token accounting reveals the cost asymmetry: baseline query tokens total 4686 with zero seed tokens, giving 234.30 tokens per question. The enhanced arm uses 3864 query tokens but 97805 seed tokens from 400 model-backed write calls, totalling 101669 tokens or 5083.45 per question. Query latency is 408.03 milliseconds for the baseline versus 503.76 milliseconds for the enhanced arm, but end-to-end latency including writes is 408.03 milliseconds for the baseline versus 7375.67 milliseconds for the enhanced arm. No records were excluded from either arm (0 baseline, 0 enhanced).

5.2 EXPANDED CONTEXT BUDGET

At the expanded budget of 2400 characters, the baseline achieves accuracy of 60.0% and evidence recall of 55.0%, with 12 correct out of 20. The enhanced rail reaches accuracy of 95.0% and evidence recall of 95.0%, with 19 correct out of 20. The accuracy gain narrows to 35 percentage points. The paired decomposition shows 12 both-correct, 7 enhanced-only, 0 baseline-only, and 1 both-wrong. The baseline improves as the larger budget exposes more of the file. In the paired outcomes, the retrieved arm answers additional questions for which the baseline lacks complete evidence; presentation and selection still vary jointly. The single both-wrong case concerns a dog-breed question where neither arm delivered the complete target fact. Notably, the baseline records 1 correct answer without complete evidence recall: a partial fact fragment still contained the accepted answer string, demonstrating that strict evidence recall is not a necessary condition for correctness under string-match scoring. The enhanced arm shows zero correct-without-full-evidence and zero wrong-with-full-evidence cases.

Wilson intervals at the expanded budget are 38.7% to 78.1% for the baseline and 76.4% to 99.1% for the enhanced arm. Token accounting shows baseline query tokens of 16877 with zero seed tokens, yielding 843.85 tokens per question. The enhanced arm uses 14638 query tokens and 249398 seed tokens from 400 write calls, totalling 264036 tokens or 13201.80 per question. Query latency is 717.01 milliseconds for the baseline versus 601.41 milliseconds for the enhanced arm; the enhanced arm has lower observed query latency at this budget, but a single run cannot establish why this difference occurred or whether it would persist. However, end-to-end latency including writes is 717.01 milliseconds for the baseline versus 7133.41 milliseconds for the enhanced arm. No records were excluded (0 baseline, 0 enhanced).

Total cost per correct answer offers a complementary, protocol-specific measure. At the restricted budget it is undefined for the baseline because that arm answers no question correctly; the enhanced arm uses 7820.69 tokens per correct answer. At the expanded budget, the corresponding values are 1406.42 for the baseline and 13896.63 for retrieved memory, a ratio of 9.88. This quantity allocates all recorded expenditure across successful answers rather than measuring the price of an individual successful query. On a per-question basis the total-cost ratios are 21.70 and 15.64 at the restricted and expanded budgets. These derived ratios are calculated from recorded totals by the

same normalization code used for the tables. They show why lower query-token counts alone do not establish lower operating cost.

Table 2: Restricted Context Budget. Costs include memory writes; missing usage is not treated as zero.

Metric	Baseline	Enhanced
Correct answers	0	13
Paired observations	20	20
Answer accuracy	0.0%	65.0%
Evidence recall	0.0%	65.0%
Query tokens (total)	4686	3864
Write tokens (total)	0	97805
Total tokens / paired observation	234.30	5083.45
Mean query latency (ms)	408.03	503.76
Amortized total latency (ms)	408.03	7375.67
Excluded records	0	0

Table 3: Expanded Context Budget. Costs include memory writes; missing usage is not treated as zero.

Metric	Baseline	Enhanced
Correct answers	12	19
Paired observations	20	20
Answer accuracy	60.0%	95.0%
Evidence recall	55.0%	95.0%
Query tokens (total)	16877	14638
Write tokens (total)	0	249398
Total tokens / paired observation	843.85	13201.80
Mean query latency (ms)	717.01	601.41
Amortized total latency (ms)	717.01	7133.41
Excluded records	0	0

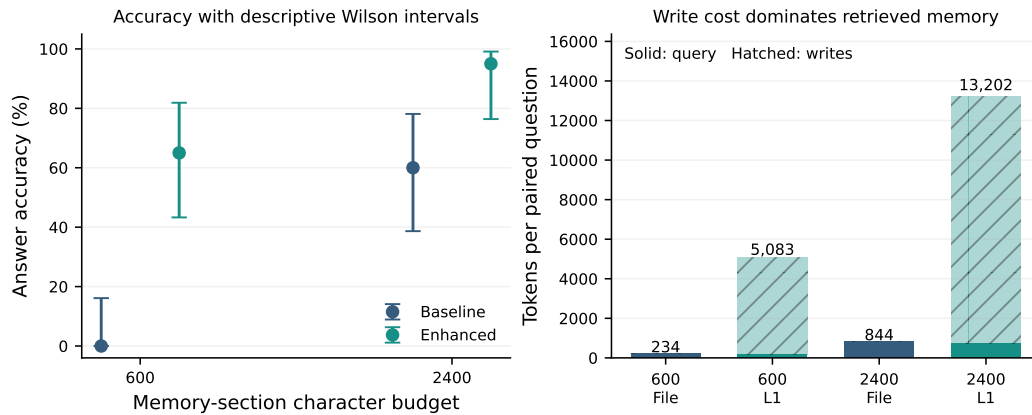


Figure 2: Quality and resource tradeoff. Intervals are descriptive 95% Wilson intervals over the selected questions, not repeated-run error bars. Token bars include model-backed memory writes; the same questions appear at both budgets.

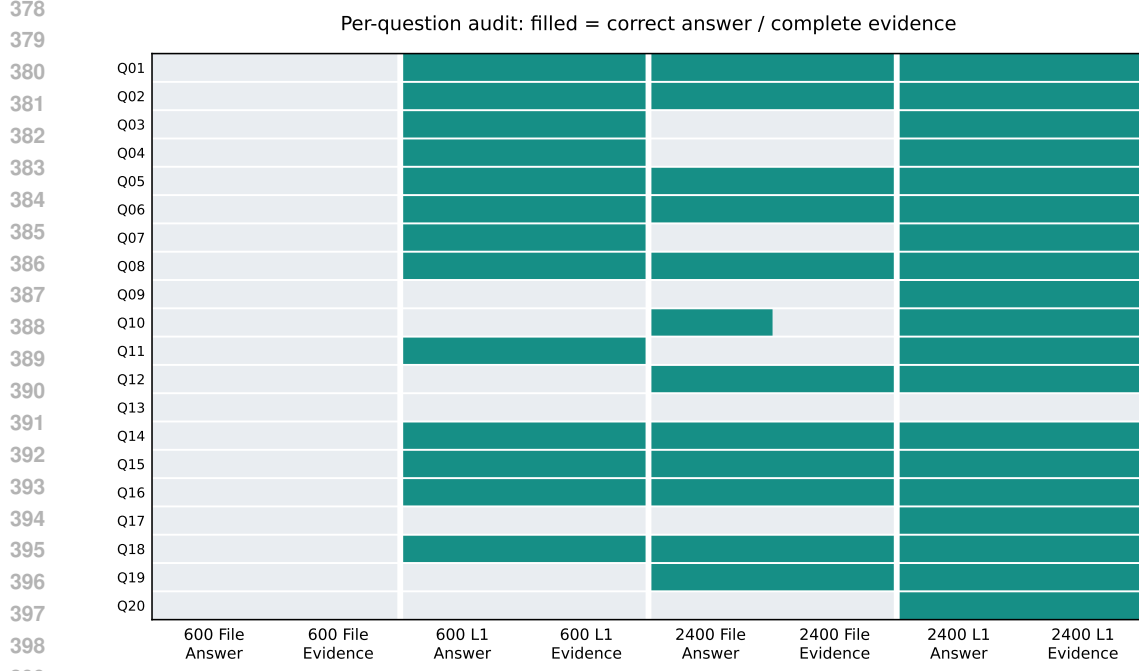


Figure 3: Matched question-level correctness and full-evidence recall across both budgets. Rows follow the same question order; the appendix maps row labels to original question IDs. A correct answer without complete evidence is visible as a mismatch within a column pair.

6 DISCUSSION AND LIMITATIONS

The observations answer a system-design question with an explicit trade-off. Retrieved memory supplies more complete target facts and yields more accepted answers under both budgets, while the file-prefix baseline avoids model-backed construction. The largest accuracy gap occurs in the condition where the baseline delivers no complete target facts. That is evidence of a restrictive information-delivery protocol, not evidence that the model cannot answer these questions when given useful context. The expanded budget improves the baseline and narrows the gap, demonstrating budget sensitivity within the tested implementation pair.

The question audit makes the aggregate comparison more interpretable. At the expanded budget, the baseline and enhanced arms are both correct on 12 questions, only the enhanced arm is correct on 7, and both are wrong on 1. There are no baseline-only successes in this run. These paired counts reveal where the marginal accuracy difference comes from without treating the two arms as unrelated samples. They do not show which ranking signal was decisive, because the stored result format does not contain per-candidate score decompositions or packing decisions.

A dog-breed question remains wrong in both arms at the expanded budget. The recorded enhanced context lacks the complete fact, making evidence delivery a plausible place to investigate; however, lexical mismatch, candidate selection, display truncation and whole-item packing remain competing explanations. At the restricted budget, several additional retrieved-arm questions also lack complete evidence. It would be incorrect to call these failures budget-independent, because some disappear at the expanded allowance. A follow-up should record candidate scores and skipped-item reasons while keeping the item representation fixed. The present records support this diagnostic agenda but cannot supply its missing measurements.

Evidence recall and correctness also capture different events. At the expanded budget, the baseline has 1 correct answer without complete-fact recall. A retained partial fact contains an accepted answer span, as illustrated in the diagnostic appendix. In the enhanced arm the two indicators coincide on every observed question, which is consistent with its whole-item presentation but does not prove that full evidence is necessary for answer generation in general. Substring-based scoring

may count an answer span in an otherwise poor response or miss an acceptable paraphrase. These cases motivate semantic review rather than treating either automatic indicator as a complete measure of response quality.

The measured write expense belongs to this construction protocol. Each question rebuilds a fresh memory episode, and the enhanced arm pays model calls while the file arm receives direct writes. An application that reuses an unchanged memory for many questions would allocate write cost differently; that workload was not measured here. Direct insertion of the same memory representation without model acknowledgements is another plausible engineering variant, but no corresponding ablation was run. The present evidence supports reporting write and query usage separately and measuring the application’s rebuild frequency before making deployment-cost claims.

Several restrictions prevent broader attribution. File metadata and prefix ordering consume usable context differently from dialogue-formatted items, so equal character caps do not equalize information or tokens. The ranking weights and item display cap were not varied. The selected single-session slice omits temporal updates, abstention and multi-session reasoning. There is one run per budget, so run-to-run variation and API timing stability are unknown. L2/L3 persistence and team synchronization were not evaluated. Format-controlled comparisons, repeated runs, score traces and the full benchmark would address distinct gaps; the present budget comparison substitutes for none of them.

7 CONCLUSION

In this audited implementation pair, retrieved L1 memory improves answer accuracy over file-prefix memory under both full-section character budgets. The observed gains are 65 percentage points at the restricted allowance and 35 percentage points at the expanded allowance. Complete-evidence delivery tracks correctness closely, with a baseline partial-evidence exception that illustrates the distinction between the two metrics. The resource conclusion runs in the opposite direction: including model-backed construction makes retrieved memory more expensive despite lower query-token totals. This is a trade-off between concrete systems, not a causal estimate of the ranking algorithm’s contribution.

The practical contribution is the joint audit of delivered evidence, answer outcomes and write-inclusive cost. Saved records, paired diagnostics and explicit accounting make both the benefit and the expense inspectable. Further work should control representation and construction, record retrieval-stage decisions, repeat the experiments and broaden the task coverage before generalizing these single-run findings. The current artifact provides a reproducible starting point for those experiments rather than a claim of benchmark leadership.

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A QUESTION-LEVEL AUDIT AND REPRODUCIBILITY

The following identifiers link directly to the submitted raw records. A and E denote answer correctness and complete-evidence recall. These are recorded single-run outcomes, not independent replications across budgets.

Table 4: Question audit at the 600-character budget.

Row	Original question ID	File A	File E	L1 A	L1 E
Q01	0862e8bf	0	0	1	1
Q02	118b2229	0	0	1	1
Q03	15745da0	0	0	1	1
Q04	19b5f2b3	0	0	1	1
Q05	1faac195	0	0	1	1
Q06	36580ce8	0	0	1	1
Q07	3d86fd0a	0	0	1	1
Q08	545bd2b5	0	0	1	1
Q09	58bf7951	0	0	0	0
Q10	60d45044	0	0	0	0
Q11	66f24dbb	0	0	1	1
Q12	6ade9755	0	0	0	0
Q13	75499fd8	0	0	0	0
Q14	86f00804	0	0	1	1
Q15	8a137a7f	0	0	1	1
Q16	8ebdbe50	0	0	1	1
Q17	c14c00dd	0	0	0	0
Q18	c8c3f81d	0	0	1	1
Q19	caf9ead2	0	0	0	0
Q20	ccb36322	0	0	0	0

Table 5: Question audit at the 2400-character budget.

Row	Original question ID	File A	File E	L1 A	L1 E
Q01	0862e8bf	1	1	1	1
Q02	118b2229	1	1	1	1
Q03	15745da0	0	0	1	1
Q04	19b5f2b3	0	0	1	1
Q05	1faac195	1	1	1	1
Q06	36580ce8	1	1	1	1
Q07	3d86fd0a	0	0	1	1
Q08	545bd2b5	1	1	1	1
Q09	58bf7951	0	0	1	1
Q10	60d45044	1	0	1	1
Q11	66f24dbb	0	0	1	1
Q12	6ade9755	1	1	1	1
Q13	75499fd8	0	0	0	0
Q14	86f00804	1	1	1	1
Q15	8a137a7f	1	1	1	1
Q16	8ebdbe50	1	1	1	1
Q17	c14c00dd	0	0	1	1
Q18	c8c3f81d	1	1	1	1
Q19	caf9ead2	1	1	1	1
Q20	ccb36322	0	0	1	1

B OBSERVED DIAGNOSTIC CASES

Cases below are selected deterministically for diagnostic categories, not as a random qualitative sample. Answers are verbatim recorded outputs; questions are copied from the adapted dataset.

Retrieved-Only Success. Budget 600; question 0862e8bf.

Question: What is the name of my cat?

Accepted answer: Luna

File answer: UNKNOWN

Retrieved answer: Your cat's name is Luna.

Retrieval Failure. Budget 600; question 58bf7951.

Question: What play did I attend at the local community theater?

Accepted answer: The Glass Menagerie

File answer: UNKNOWN

Retrieved answer: UNKNOWN

Partial Evidence. Budget 2400; question 60d45044.

Question: What type of rice is my favorite?

Accepted answer: Japanese short-grain rice

File answer: Your favorite type of rice is Japanese short-grain rice.

Retrieved answer: Your favorite type of rice is Japanese short-grain rice.

C ARTIFACT AND GENERATION PROVENANCE

The submission includes the adapted data, original write/query records, normalization code, citation metadata and a hashed source snapshot. Writing assistance uses a separate model from the experimental query model. Its plan, draft, critique, revision and usage are saved separately; none of those calls contributes to the experimental accuracy. The methodology schematic is AI-generated and checked against the implementation; all empirical plots and tables are rendered programmatically from the recorded data.